

Mode-Converted Specular Reflections and Evanescent Incidence in the Periodic Slab Solver

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1 Introduction and scope

This document records three connected developments, each validated against an independent exact reference.

1. **Five-channel Mie verification (sphere).** The elastic Mie solution and the Foldy–Lax voxelised sphere were compared phase-sensitively (real and imaginary parts separately) across all five non-zero channels of the 3×3 scattering matrix ($P \rightarrow P$, $P \rightarrow SV$, $SV \rightarrow P$, $SV \rightarrow SV$, $SH \rightarrow SH$) at $ka = 0.1$, 0.5 , and 1.5 .
2. **Full specular reflection matrix of the periodic heterogeneous slab.** The Weyl lattice-sum extraction, previously limited to R_{PP} , was generalised to the 2×2 P–SV matrix plus the scalar SH channel at arbitrary horizontal slowness p , and validated channel-by-channel against Kennett reflectivity. The ocean-bottom study now propagates the full 2×2 sub-ocean reflectivity, so that internal $P \rightarrow S \rightarrow P$ conversion contributes to the observable water-column R_{PP} .
3. **Physical evanescent incidence.** Incident fields are constructed from complex slowness vectors, so post-critical incidence ($p > 1/\alpha$) is represented by the true exponentially decaying inhomogeneous wave rather than a homogeneous-wave surrogate.

Two pre-existing defects were found and repaired along the way (Section 3.2), and two normalisation conventions were pinned by measurement rather than assumption (Sections 2.1 and 3.3). All numerical values quoted below are reproduced by the committed test suite (641 tests at the time of writing).

Throughout, coordinates follow the package convention: z is axis 0 (positive down), x is axis 1, y is axis 2; the sagittal plane is z – x . Vertical slownesses are $\eta_m = \sqrt{1/c_m^2 - p^2}$ with the branch $\text{Im } \eta_m \geq 0$, and all complex polarisation algebra uses analytic continuation (no conjugation).

2 Five-channel Mie verification for the sphere

For an isotropic sphere the P–SV and SH problems decouple at each partial-wave order, leaving five non-zero channels of the scattering matrix. P -wave incidence is axially symmetric ($m = 0$);

SV and SH plane-wave incidence carry azimuthal order $m = 1$, with far-field angular functions $P_n^1(\cos \theta) = dP_n/d\theta$ (Condon–Shortley), $dP_n^1/d\theta$, and $P_n^1/\sin \theta$ respectively.

2.1 The $m=1$ renormalisation

The scattering coefficients are stored in the $m=0$ plane-wave expansion normalisation. Evaluating SV/SH far fields requires the $m=1$ expansion coefficient of $\hat{\mathbf{x}} e^{ik_s z}$, which differs by the measured factor

$$\text{renorm}(n) = -\frac{1}{n(n+1)}, \quad (1)$$

the minus sign arising from the $\nabla \times \nabla \times (\mathbf{r} \psi)$ convention. (The planning estimate $+i/n(n+1)$ was wrong; the correct factor was pinned by the Rayleigh-limit comparison and confirmed by reciprocity.) A further empirical subtlety: the $SH \rightarrow SH$ far field at $\varphi = 0$ is carried by the N-type (poloidal, b_n^{SV}) coefficients; the true M-type (c_n) contribution is $\mathcal{O}((ka)^4)$ and negligible.

2.2 Reciprocity and results

The off-diagonal channels obey

$$k_P^2 f_{PS}(\theta) = -k_S^2 f_{SP}(\theta), \quad (2)$$

with the conversion constant exactly -1 (measured ratio $-1.0002 + 0.0002i$ at $ka = 0.05$, -1.0000 at $ka = 0.5$; asserted in `TestReciprocity`). Phase-sensitive errors between Mie and the voxelised Foldy–Lax sphere (relative to the pattern peak; Im errors were ≈ 0 in every case):

Channel	$ka = 0.1$ ($n_{\text{sub}}=4$)	$ka = 0.5$ ($n_{\text{sub}}=6$)	$ka = 1.5$ ($n_{\text{sub}}=8$)
$P \rightarrow P$	0.041	0.200	0.316
$P \rightarrow SV$	0.038	0.202	0.393
$SV \rightarrow P$	0.048	0.186	0.368
$SV \rightarrow SV$	0.047	0.228	0.565
$SH \rightarrow SH$	0.047	0.229	0.573

The errors scale as $\mathcal{O}(ka)$ (voxelisation and T-matrix truncation at finite contrast), falling consistently under mesh refinement. The identical Mathematica pipeline (`FiveChannelExtension.wl`, run headlessly by `FiveChannelDriver.wl`) reproduces all five channels to better than 10% at $k_S a = 0.52$ with 9 voxels per diameter, with the SV/SH channel errors agreeing to ten digits — a free symmetry check of the two independent solves.

3 Specular reflection matrix of the periodic slab

3.1 Generalised Weyl extraction

For the infinite periodic slab the 2D lattice sum replaces the spherical-wave kernel by its specular Weyl term, $e^{ikr}/4\pi r \rightarrow i e^{i\omega\eta_m|z|}/(2\omega\eta_m d^2)$ per mode. With horizontally layer-averaged Foldy–

Lax sources $(\mathbf{f}_\ell, \boldsymbol{\sigma}_\ell)$ and the upgoing slowness vector $\mathbf{s}_m = (-\eta_m, p, 0)$, complex-unit direction $\hat{\mathbf{d}}_m = c_m \mathbf{s}_m$, the outgoing amplitudes are

$$Q_{P,\ell} = -\hat{\mathbf{d}}_P \cdot \mathbf{f}_\ell - i\omega \mathbf{s}_P \cdot \boldsymbol{\sigma}_\ell \cdot \hat{\mathbf{d}}_P, \quad R_P = \frac{-i}{2\omega\eta_P d^2 \rho \alpha^2} \sum_\ell Q_{P,\ell} e^{i\omega\eta_P z_\ell}, \quad (3)$$

$$\mathbf{Q}_{S,\ell} = -\mathbf{f}_\ell - i\omega \boldsymbol{\sigma}_\ell \cdot \mathbf{s}_S, \quad R_{SV,SH} = \frac{-i}{2\omega\eta_S d^2 \rho \beta^2} \sum_\ell (\hat{\mathbf{e}}_{SV}, \hat{\mathbf{e}}_{SH}) \cdot \mathbf{Q}_{S,\ell} e^{i\omega\eta_S z_\ell}, \quad (4)$$

with $\hat{\mathbf{e}}_{SV} = \beta(p, \eta_S, 0)$ and $\hat{\mathbf{e}}_{SH} = (0, 0, 1)$. Only the force term is negated (the T-matrix convention $+V\omega^2 \Delta\rho^* \mathbf{u}$ is opposite to the Lippmann–Schwinger body force). The full reflection matrix is assembled from three periodic solves (incident P , SV , SH at the same slowness), sharing the local T-matrices.

3.2 Two pre-existing defects

Oblique stress coupling. The original R_{PP} extraction used the vertical wavenumber in the stress term, $-ik_z(\hat{\mathbf{d}} \cdot \boldsymbol{\sigma} \cdot \hat{\mathbf{d}})$ with $k_z = \omega\eta_P$. The correct coupling is the full wave vector, $-i\omega(\mathbf{s} \cdot \boldsymbol{\sigma} \cdot \hat{\mathbf{d}}) = -ik_P(\hat{\mathbf{d}} \cdot \boldsymbol{\sigma} \cdot \hat{\mathbf{d}})$. The two coincide only at $p = 0$ — and no oblique slab-versus-Kennett test existed, so the defect had never been observed. Equation (3) is the corrected form; the oblique validation of Table 1 is its proof.

$p \rightarrow 0$ polarisation discontinuity. The incident-field builder’s vertical-incidence special case selected the SV polarisation $+\hat{\mathbf{x}}$, whereas the $p \rightarrow 0^+$ limit of its own generic branch is $-\hat{\mathbf{x}}$. The discontinuity inverted the sign of R_{SS} at exactly $p = 0$, made observable by Kennett’s convention $K_{SS}(0) = -K_{SH}(0)$ (the SV sign flips under reflection; same physics, opposite bookkeeping to SH). The special case now takes the continuous limit.

3.3 The convention bridge to Kennett

The slab extraction produces displacement-amplitude ratios; the package’s Kennett implementation stores reflection matrices in its modified (eigenvector-normalised) convention, in which they are symmetric. The planned conversion $\tilde{R}_{ij} = \sqrt{\eta_i/\eta_j} R_{ij}$ proved wrong by measurement. The correct bridge is the diagonal similarity

$$\tilde{\mathbf{R}} = \mathbf{D} \mathbf{R} \mathbf{D}^{-1}, \quad \mathbf{D} = \text{diag}(\alpha\sqrt{\eta_P}, i\beta\sqrt{\eta_S}), \quad (5)$$

the velocity factors and the factor i on SV reflecting Kennett’s eigenvector normalisation (the $-2i$ factor in the off-diagonal construction of `psv_solid_solid`). The decisive, convention-independent check is the reciprocity invariant: the product of the two off-diagonal slab/Kennett ratios is invariant under *any* diagonal similarity and measured $0.987 \approx 1$, proving the physics correct before the bookkeeping was; the per-channel ratios under (5) are then real and within 1% of unity. Diagonal entries are unchanged by any diagonal similarity, so the historical R_{PP} results are unaffected.

3.4 Channel validation

Table 1. Uniform slab versus Kennett (modified convention). Background $\alpha = 5000$, $\beta = 3000$ m/s, $\rho = 2500$ kg/m³; contrast $\Delta\lambda = 2$, $\Delta\mu = 1$ GPa, $\Delta\rho = 100$ kg/m³; $\omega = 150$ rad/s; $H = 4$ m, mesh $a = 2$ m, $N_z = 1$ unless stated. Relative errors.

Channel	$p = 0$	$p = 1 \times 10^{-4}$ s/m	$p = 2.5 \times 10^{-4}$ s/m (post-critical)
R_{PP}	0.0045	0.0048	0.054 (0.0009 at $a=0.5$, $N_z=4$)
R_{PS}	(vanishes; $< 10^{-2}$ rel.)	0.0066	0.033
R_{SP}	(vanishes)	0.0066	0.033
R_{SS}	0.0032	0.0110	0.043
R_{SH}	0.0032	0.0057	0.044

The modified matrix is symmetric to 1.3×10^{-5} at oblique incidence (independent solves: R_{SV} from P incidence versus R_P from SV incidence), the conversion channel is Born-linear at weak contrast (doubling ratio 2.0, rtol 0.05), and at $p = 0$ the degeneracy $R_{SH} = -\tilde{R}_{SS}$ holds at machine level (the SV and SH systems on the square lattice are exact 90° rotations of one another).

4 Ocean-bottom embedding: the 2×2 sub-ocean recursion

The ocean-bottom study (water | heterogeneous sediment slab | elastic halfspace) previously injected only the scalar R_{PP} of the slab into a scalar water-step recursion. Internal mode conversion — P converting to SV inside the sediment package, reverberating, and converting back — was absent from the observable. The sub-ocean reflectivity is now the full matrix

$$\mathbf{M}(\omega) = \mathbf{E} \mathbf{R}^{\text{sed|hs}} \mathbf{E} + \tilde{\mathbf{R}}^{\text{slab}}(\omega), \quad \mathbf{E} = \text{diag}(e^{i\omega\eta_P^{\text{sed}}H}, e^{i\omega\eta_S^{\text{sed}}H}), \quad (6)$$

so a conversion path acquires the mode-correct two-way phase $E_i E_j$, and the water-side observable is

$$R_{PP}^{\text{obs}} = \left[\mathbf{R}_d + \mathbf{T}_u \mathbf{M} (\mathbf{I} - \mathbf{R}_u \mathbf{M})^{-1} \mathbf{T}_d \right]_{00}, \quad (7)$$

with the fluid–solid coefficients of `psv_fluid_solid`. SH cannot couple through the fluid and is omitted (the slab-level R_{SH} remains available directly). Per-frequency cost is two Foldy–Lax solves (P - and SV -incident).

Anchors. At zero contrast the decomposed path reproduces the full-stack Kennett R_{PP} to relative error 2×10^{-16} at both $p = 0$ and $p = 2 \times 10^{-4}$ s/m (asserted at rtol 10^{-10}). The oblique case is the load-bearing anchor: at $p = 0$ every quantity introduced by the matrix upgrade is invisible, and a deliberately injected phase error of the $E_i E_j$ type produces a 133% error that only the oblique anchor detects.

Magnitude of the restored physics. At moderate contrast and oblique incidence, suppressing the off-diagonal entries of $\mathbf{M}$ shifts the observable R_{PP} at the $\sim 10\%$ level — dominated by the background sediment/halfspace interface conversion, which the old scalar recursion also lacked. The slab’s own conversion contribution is second order in the contrast, isolated by

subtracting only $\tilde{\mathbf{R}}^{\text{slab}}$ off-diagonals: 1.2×10^{-5} relative at weak contrast. (An instructive trap: zeroing the off-diagonals of the *total* $\mathbf{M}$ measures the contrast-independent background conversion, not the slab’s — the committed tests separate the two.)

5 Evanescent incidence

Past the critical slowness ($p > 1/\alpha$ for P) the incident field is an inhomogeneous wave: with $\eta_P = i|\eta_P|$,

$$\mathbf{u}^{\text{inc}}(\mathbf{x}) = \mathbf{p} e^{i\omega \mathbf{s} \cdot \mathbf{x}} = \mathbf{p} e^{i\omega p x} e^{-\omega |\eta_P| z}, \quad \mathbf{s} = (\eta_P, p, 0), \quad \mathbf{p} = \alpha \mathbf{s}, \quad (8)$$

decaying with depth. The incident-field builder originally degenerated to a horizontally propagating homogeneous wave there (constant amplitude at all depths), and the P -incident column of the reflection matrix carried a documented warning. The builder now constructs fields directly from complex slowness vectors and polarisations ($\mathbf{p} \cdot \mathbf{p} = 1$ by analytic continuation — the same convention as Kennett, though the Euclidean amplitude exceeds unity post-critically); strains follow as $\varepsilon_{ij} = \frac{i\omega}{2}(s_i p_j + s_j p_i)$. The per-layer decay ratio matches $e^{-\omega |\eta_P| \Delta z}$ to 10^{-10} , and the sub-critical path reproduces the previous builder to 2.6×10^{-16} .

Channel ($p = 2.5 \times 10^{-4}$ s/m)	homogeneous surrogate	evanescent field
R_{PP}	0.637	0.054
R_{PS}	0.965	0.033
R_{SP}	0.033	0.033

Accuracy is now symmetric across the critical slowness (errors 0.025 at $p = 1.9 \times 10^{-4}$ versus 0.031 at 2.1×10^{-4}), and the residual post-critical error is pure vertical discretisation: a monotonic exponential sampled by one-point quadrature biases the layer average (unlike oscillatory sub-critical phase, which self-cancels), and the error collapses $0.054 \rightarrow 0.009 \rightarrow 0.0009$ under refinement $(a, N_z) = (2, 1) \rightarrow (1, 2) \rightarrow (0.5, 4)$ at fixed H — pinned by a committed convergence test. The conversion channels floor near 2%, the known Rayleigh T-matrix truncation scale at this contrast. The only remaining special points are the exact grazing slownesses $p = 1/\alpha, 1/\beta$, where the Weyl prefactor $1/\eta$ is genuinely singular.

6 Code artifacts

Artifact	Role
<code>sphere_scattering.py</code>	five-channel Mie far fields, SV/SH decomposition
<code>FiveChannelExtension.wl</code> , <code>FiveChannelDriver.wl</code>	Mathematica five-channel comparison (headless)
<code>slab_weyl_amplitudes</code>	shared $P/SV/SH$ Weyl extractor, eqs. (3)–(4)
<code>SlabReflectionMatrix.to_modified</code>	convention bridge, eq. (5)
<code>slab_reflection_matrix</code>	three-solve matrix assembly at slowness p
<code>kennett_reference_matrix</code>	five-channel Kennett reference for a uniform layer
<code>_build_slab_incident_field_slowness</code>	complex-slowness (evanescent) incident fields
<code>_kennett_water_step</code> (2×2), <code>MT_psv</code>	ocean-bottom sub-ocean matrix recursion

All quoted measurements are asserted by the test suite (`TestCompleteScatteringMatrix`, `TestSlabReflectionMatrix`, `TestEvanescentIncidence`, `TestModeConvertedOceanBottom`, and the oblique zero-contrast anchors), 641 tests passing at the time of writing.